

Latent Structure Discovery via PCA, ICA, RP, KMeans, and EM

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“All models are wrong, but some are useful.”
~ George E. P. Box

Abstract—This report evaluates five unsupervised learning methods, K-Means, Expectation Maximization (EM), PCA, ICA, and Random Projection (RP), on two real-world datasets: Global Cancer Patients and Company Bankruptcy. We examine the impact of dimensionality reduction on clustering quality and neural network (MLPClassifier) performance. We hypothesize that reducing feature dimensionality will improve clustering by removing noise and enable faster, equally effective neural network training. Results confirm this: clustering metrics generally improved post-reduction, and MLP performance remained stable or improved, with reduced training time. RP provided the best trade-off between speed and accuracy.

I. INTRODUCTION

Unsupervised learning enables the discovery of hidden structure in unlabeled data, making it essential in domains where annotated labels are scarce or costly [3]. This paper investigates two major components of unsupervised learning: clustering and dimensionality reduction. Specifically, we evaluate two clustering algorithms, K-Means and Expectation Maximization (EM), and three linear dimensionality reduction techniques, Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection (RP).

K-Means partitions data by assigning each point to its nearest cluster centroid, iteratively refining centroids to minimize within-cluster variance. EM, implemented via Gaussian Mixture Models, performs soft clustering by assigning probabilistic memberships to each cluster. These algorithms, particularly K-Means, are sensitive to the curse of dimensionality, where distance metrics lose discriminative power in high-dimensional spaces. Dimensionality reduction addresses this by projecting data into a more compact space, ideally preserving essential structure. PCA identifies axes of maximum variance, ICA separates statistically independent non-Gaussian signals, and RP uses randomized linear projections to approximately preserve pairwise distances.

We apply these techniques to two real-world datasets from Kaggle: the *Global Cancer Patient Dataset*, which contains high-dimensional mixed categorical and numerical attributes, and the *Company Bankruptcy Dataset*, which is numerical, high-dimensional, and class-imbalanced. Both present challenges for clustering and downstream supervised learning due to sparsity, redundancy, and skewed class distributions.

This study has two main objectives: (1) to evaluate how dimensionality reduction impacts clustering performance; and (2) to assess its influence on the efficiency and generalization of a Multi-Layer Perceptron (MLP) classifier. We hypothesize that dimensionality reduction will improve clustering by

mitigating noise and redundancy, and will accelerate neural network training without significantly degrading classification accuracy. These hypotheses are tested using silhouette scores and Adjusted Rand Index (ARI) for clustering, and F1-score, accuracy, and training time for supervised learning.

II. METHODOLOGY

This experiment was designed to investigate how unsupervised learning techniques can uncover latent structure in high-dimensional datasets and how dimensionality reduction impacts clustering quality and neural network performance. We evaluated five unsupervised algorithms, K-Means, Expectation Maximization (via Gaussian Mixture Models), Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection (RP), on two publicly available datasets from Kaggle: the *Global Cancer Patients Dataset* and the *Company Bankruptcy Dataset*.

A. Datasets

Dataset 1: Global Cancer Patients dataset includes clinical and demographic records from cancer patients worldwide between 2015 and 2024. It contains both numerical and categorical attributes. Initial preprocessing involved removing the `Patient_ID` column, which served only as a unique identifier. Four categorical variables, Gender, Country Region, Cancer Type, and Cancer Stage, were transformed via one-hot encoding, increasing the dimensionality from 15 original features to 34 total features after encoding. Numerical features were standardized using z-score normalization to ensure all features contributed equally during distance-based clustering.

This preprocessing was critical for clustering algorithms like K-Means and EM, which rely on distance metrics. Without scaling, features with larger ranges would disproportionately influence clustering outcomes.

Dataset 2: Company Bankruptcy dataset includes financial indicators for thousands of companies, with a binary label indicating whether the company filed for bankruptcy. It contains 6,819 samples and 96 numerical features, all of which were retained. As all features were numeric, one-hot encoding was not required. However, standardization was again applied to place all features on a common scale.

This dataset is also highly imbalanced, with approximately 97% of instances labeled as non-bankrupt and only 3% as bankrupt. This class imbalance presents a challenge for supervised learning, especially neural networks, which may overfit to the majority class without proper preprocessing

or resampling techniques. For this reason, we selected this dataset for evaluating the MLP classifier before and after dimensionality reduction.

B. Supervised Learning Setup

For the supervised component of this study, we refit a Multi-Layer Perceptron (MLP) classifier from the prior assignment using the Company Bankruptcy dataset. The goal was to assess the impact of dimensionality reduction on model training time, predictive accuracy, and generalization performance. We compared the baseline (raw data) model to those trained on PCA-, ICA-, and RP-transformed feature spaces.

C. Algorithms Used

All algorithms in this experiment were implemented using the `scikit-learn` library. This included clustering algorithms such as KMeans and Expectation Maximization (via `GaussianMixture`), as well as dimensionality reduction methods: Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection (RP). The Multi-Layer Perceptron (MLP) classifier was also utilized for the supervised learning portion of the study.

Default hyperparameters were used unless otherwise specified. Parameter tuning details, such as the choice of number of clusters or components, initialization strategy, or iteration limits, are discussed in the corresponding results and evaluation sections for each algorithm.

D. Evaluation Metrics and Component Selection

To evaluate the quality of clustering, we used Silhouette Score. This metric was computed across a range of cluster values ($k \in [2, 14]$) to identify trends and support algorithm comparisons without relying on ground truth labels.

For dimensionality reduction, we employed method-specific strategies to select the optimal number of components. PCA used the cumulative explained variance curve to retain at least 95% of the variance. ICA relied on the absolute average kurtosis across components to identify statistically independent, non-Gaussian sources. Random Projection (RP) used reconstruction error, computed as the mean squared error (MSE) between the original and reconstructed data via the pseudo-inverse of the projection matrix, to guide selection.

Once optimal component counts were determined, clustering algorithms were re-applied to the reduced datasets to evaluate improvements in structure. A similar approach was used for the MLP classifier, where models were trained on transformed data using each reduction method. Performance was assessed using accuracy, F1-score, and training time.

III. RESULTS

A. KMeans & Expectation Maximization Clustering

KMeans clustering is a non-parametric, centroid-based unsupervised machine learning algorithm. It partitions the dataset into k clusters by iteratively assigning each data point to the nearest cluster centroid and updating the centroids as

the mean of their assigned points. This process continues until convergence, typically when centroid positions stabilize or cluster memberships no longer change.

a) *Global Cancer Patients Dataset*: Two key parameters were tuned for : the number of clusters (`n_clusters`) and the number of initializations (`n_init`). The parameter `n_init` controls how many times the algorithm is run with different centroid seeds, and the best output (inertia-wise) is selected. We tested `n_init` values of 5, 10, and 15 and found that `n_init = 10` provided stable results, which were used consistently throughout the experiments.

We evaluated clustering performance for both KMeans and Expectation Maximization (EM, via `Gaussian Mixture Models`) across k values from 2 to 14. Figures 1 show the results using Silhouette Score and Davies-Bouldin Index (DBI), respectively.

Figure 1 shows that KMeans outperforms EM across all cluster sizes, achieving silhouette scores between 0.05 and 0.06. The highest silhouette score for KMeans occurs at $k = 2$, after which the scores plateau, suggesting weak but stable clustering structure in the raw high-dimensional space. EM performs significantly worse, with silhouette scores consistently below 0.01, indicating poor cohesion and separation of clusters.

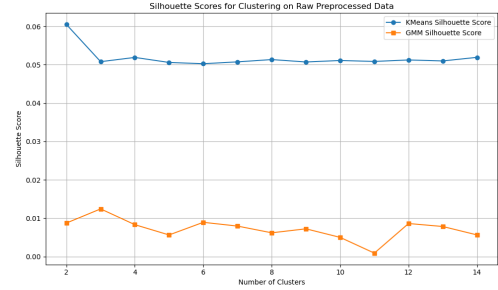


Fig. 1. Silhouette Scores for KMeans and EM on Raw Preprocessed Data

These findings support our hypothesis that the raw feature space lacks strong natural clusters, particularly for soft clustering methods like EM, and that dimensionality reduction may help uncover more meaningful structure.

b) *Company Bankruptcy*: The same clustering algorithms, KMeans and Expectation Maximization (EM), were applied to the Company Bankruptcy dataset using a slightly extended cluster range of $k \in [2, 49]$. We evaluated clustering performance using Silhouette Score and Davies-Bouldin Index (DBI).

Figure 2 shows that KMeans consistently outperforms GMM across all cluster sizes, with the highest silhouette score at $k = 2$. This suggests the dataset contains limited natural cluster structure in its raw form. Due to the high dimensionality and class imbalance (97% non-bankrupt), distance metrics may fail to reveal separable groupings. We fix $k = 2$ as the baseline cluster count for downstream DR experiments. GMM performance remained consistently lower, likely due to weak separation and the softness of its cluster boundaries. The results again support our hypothesis:

clustering quality is poor in the unreduced feature space due to high dimensionality and class imbalance.

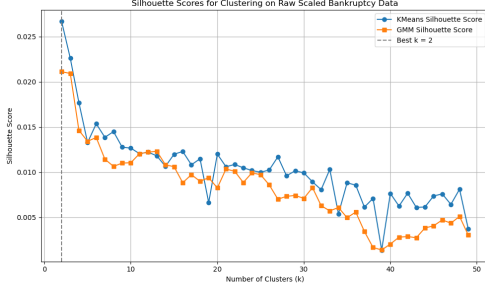


Fig. 2. Silhouette Scores for KMeans and GMM on the Company Bankruptcy Dataset.

Clustering performance varied due to differences in data structure. The Cancer dataset, with mixed types and one-hot encoding, resulted in a sparse, high-dimensional space where KMeans performed modestly but GMM struggled with non-Gaussian distributions. The Bankruptcy dataset, though fully numeric, suffered from extreme class imbalance, causing both methods, especially GMM, to overfit to the majority class. These patterns highlight how feature encoding and class distribution impact clustering effectiveness.

B. Principal Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality reduction technique that projects data into a lower-dimensional space by creating orthogonal components that maximize variance. It is commonly used to eliminate redundant features while preserving as much variability in the dataset as possible.

a) *Global Cancer Patients Dataset*: For the Global Cancer Patients dataset, we used PCA to determine the number of components required to retain 95% of the total variance. Figure 3 shows the cumulative explained variance curve. Exactly 24 components were required to retain 95% of the original variance, out of 34 initial features. This indicates substantial redundancy in the data and suggests that a compressed 24-dimensional representation is sufficient for most downstream analysis. This supports our hypothesis that dimensionality reduction can preserve key data characteristics while removing noise and redundancy.

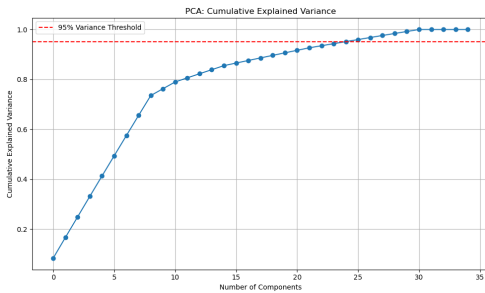


Fig. 3. Cumulative explained variance using PCA on the Global Cancer Patients dataset.

b) *Company Bankruptcy Dataset*: Similarly, PCA was applied to the Company Bankruptcy dataset. As shown in Figure 4, exactly 50 principal components were required to retain 95% of the variance out of the original 96 features. This implies that nearly half of the features are redundant or low-variance, and can be discarded without significant information loss. Such dimensionality reduction is expected to improve the efficiency of clustering and supervised models by reducing computational cost and overfitting risk. This finding supports our hypothesis that dimensionality reduction helps compress data while preserving predictive signal.

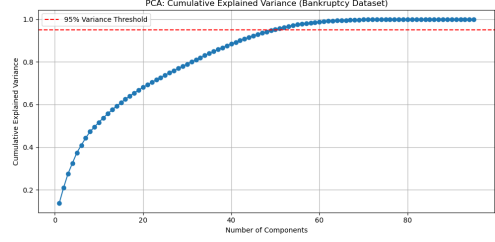


Fig. 4. Cumulative explained variance using PCA on the Company Bankruptcy dataset.

C. Independent Component Analysis

a) *Global Cancer Patients Dataset*: Independent Component Analysis (ICA) is a dimensionality reduction technique that transforms the original feature space into a new set of statistically independent components. Unlike PCA, which focuses on maximizing variance, ICA identifies non-Gaussian signals by maximizing statistical independence, often measured through kurtosis. Components with high absolute kurtosis values are considered more non-Gaussian and, therefore, more informative.

Figure 5 shows the absolute kurtosis of ICA components for the Global Cancer Patients dataset. Most components fall between kurtosis values of 0.0 and 2.1, indicating a lack of strong non-Gaussian structure. As a result, ICA did not appear to extract meaningful independent signals from this dataset. This finding does not support our hypothesis, as ICA failed to provide a more compact or informative representation.

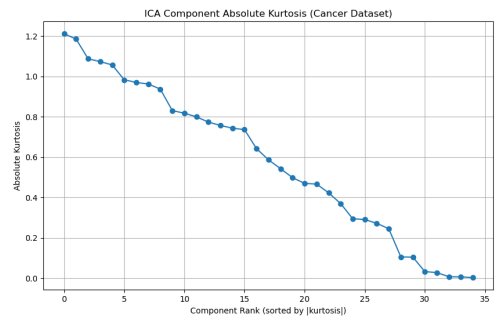


Fig. 5. ICA Component Absolute Kurtosis (Cancer Dataset)

b) Company Bankruptcy Dataset: Figure 6 shows the same plot for the Company Bankruptcy dataset. In contrast to the Cancer dataset, several components exhibit extremely high absolute kurtosis values (exceeding 5000), indicating heavy-tailed, non-Gaussian distributions. While such values may suggest strong independence, they can also arise from data sparsity or extreme outliers. For this reason, we conservatively selected the top 25 components for downstream analysis to balance informativeness and robustness. This result supports our hypothesis, as ICA successfully revealed latent structure that may benefit subsequent clustering and classification tasks.

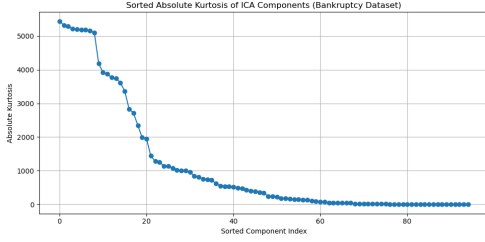


Fig. 6. Sorted Absolute Kurtosis of ICA Components (Bankruptcy Dataset)

D. Random Projection

a) Global Cancer Patients Dataset: Random Projection (RP) allows data to be randomly projected from a high-dimensional to a lower-dimensional space while approximately preserving pairwise distances. We evaluated its effectiveness using reconstruction error (mean squared error, MSE) between the original and reconstructed data using the pseudo-inverse of the projection matrix. Figure 7 shows the MSE trend as a function of the number of components for the Global Cancer Patients dataset. The error decreases rapidly at first and begins to flatten around 28 components, suggesting that most of the meaningful structure in the data can be captured in a significantly reduced subspace. This result supports our hypothesis: RP enables efficient dimensionality reduction with minimal information loss, offering a more compact representation for clustering and classification.

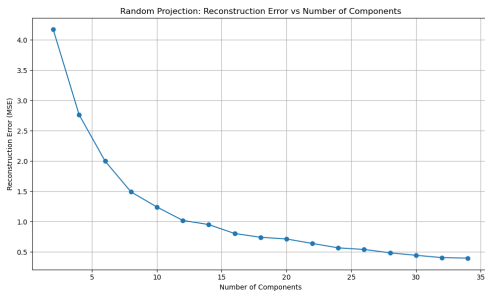


Fig. 7. Random Projection: Reconstruction Error vs Number of Components (Cancer Dataset)

b) Company Bankruptcy Dataset: Figure 8 presents the reconstruction error for the Bankruptcy dataset. Unlike the Cancer dataset, the error curve here declines in a nearly

linear fashion, from approximately 0.97 down to near-zero. This pattern suggests the dataset lacks dominant directions or feature redundancy, and that information is distributed fairly evenly across many dimensions. The absence of a distinct elbow point implies that dimensionality reduction will be less effective. Nonetheless, the smooth decline in error does allow a trade-off between dimensionality and error. This result partially supports our hypothesis: while RP can still reduce dimensionality, the Bankruptcy dataset's structure limits how aggressively this can be done without significant information loss.

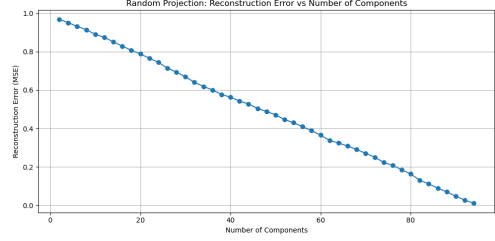


Fig. 8. Random Projection: Reconstruction Error vs Number of Components (Bankruptcy Dataset)

Each dimensionality reduction method reshapes the data differently, with implications for clustering and classification. PCA preserves global variance but assumes linearity and orthogonal components, making it ideal for eliminating redundancy and improving computational efficiency, especially in dense datasets like Cancer. ICA, in contrast, targets statistical independence and is effective when meaningful latent sources exist, as seen in the Bankruptcy dataset where it captured strong non-Gaussian signals. However, ICA is sensitive to outliers and unstable in sparse or categorical-heavy data. RP projects data using random matrices, preserving pairwise distances in expectation, which benefits KMeans and MLP training speed but may distort structure for density-based models like GMM. These underlying assumptions help explain why DR improved performance in some cases but not others.

E. KMeans and GMM Clustering with PCA

After applying PCA to each dataset, the number of components required to capture 95% of the total variance was approximately 24 for the Global Cancer Patients Dataset and 50 for the Company Bankruptcy Dataset. These transformed features were then used to re-run both KMeans and GMM clustering to evaluate the impact of dimensionality reduction on clustering performance.

a) Global Cancer Patients Dataset: As shown in Figure 9, KMeans clustering performance improved slightly after PCA, with higher silhouette scores across most values of k . This indicates that PCA reduced redundancy and noise, making clusters more distinct in the transformed feature space. In contrast, GMM initially performed slightly better at $k = 2$ before PCA but deteriorated quickly as k increased. After PCA, GMM performance worsened across nearly all

k values, suggesting that the Gaussian assumption may not hold well in the orthogonal PCA space. This partially supports our hypothesis: PCA improved KMeans clustering by simplifying the data representation, but reduced the effectiveness of soft clustering with GMM.

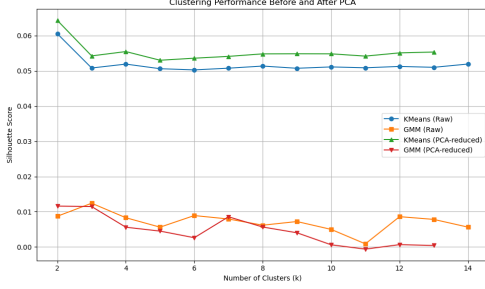


Fig. 9. Clustering Performance Before and After PCA (Global Cancer Dataset)

b) Company Bankruptcy Dataset: Figure 10 presents a similar analysis for the Company Bankruptcy dataset. Here, KMeans clustering benefited consistently from PCA across all k values. Silhouette scores were higher post-PCA, particularly for $k \geq 4$, indicating improved separation and cohesion of clusters in the reduced space. Conversely, GMM performed better in the raw high-dimensional space at lower k values but became erratic and degraded in performance after PCA, especially at higher cluster counts. This supports our hypothesis: PCA improved clustering performance for KMeans by compressing the data into a more informative subspace, but diminished GMM's performance, likely due to the loss of subtle density patterns necessary for soft clustering.

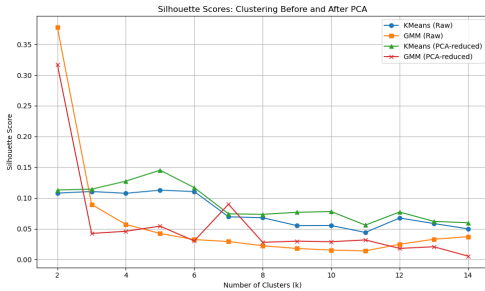


Fig. 10. Silhouette Scores: Clustering Before and After PCA (Bankruptcy Dataset)

F. KMeans and GMM Clustering with ICA

After applying ICA, we selected the number of components for dimensionality reduction based on the absolute kurtosis of the components. For the Global Cancer Patients Dataset, we retained 18 components, as the absolute kurtosis dropped below 0.5 beyond this point, indicating diminishing non-Gaussianity. For the Company Bankruptcy Dataset, we selected 48 components, where the kurtosis curve flattened, suggesting the remaining components contribute little additional independent structure.

a) Global Cancer Patients Dataset: Figure 11 shows the effect of ICA on clustering performance for the cancer dataset. Both KMeans and GMM exhibited significant improvement in silhouette scores after dimensionality reduction. The most notable gain occurred at $k = 5$, where ICA-enhanced features enabled better separation. GMM, in particular, showed the largest relative improvement, indicating that ICA uncovered latent structure better aligned with its probabilistic assumptions. This result supports our hypothesis, as ICA reduced noise and improved clustering performance.

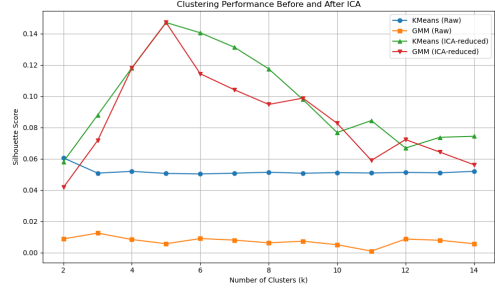


Fig. 11. Silhouette Scores: Clustering Before and After ICA (Cancer Dataset)

b) Company Bankruptcy Dataset: Figure 12 presents clustering performance after ICA for the bankruptcy dataset. KMeans showed stronger performance for smaller values of k , particularly around $k = 4$ and $k = 5$, but its effectiveness declined with increasing clusters. GMM showed minor gains at lower k but performed worse than its raw counterpart at higher values. This behavior suggests that ICA captures useful independent structure in low-cluster regimes but may lose important variance required for modeling higher- k clusters. This partially supports our hypothesis: ICA improved clustering when meaningful structure was present but hindered performance when it removed subtle variations in high-dimensional space.

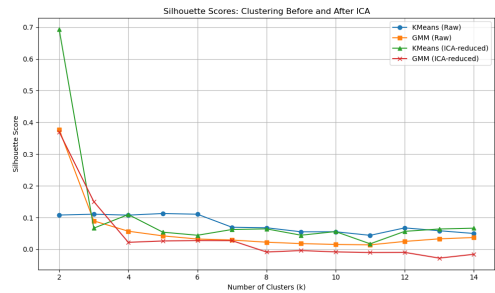


Fig. 12. Silhouette Scores: Clustering Before and After ICA (Bankruptcy Dataset)

G. KMeans and GMM Clustering with RP

After applying Random Projection (RP) to both datasets, the number of components used was chosen based on the reconstruction error (MSE) curve. For the Global Cancer Patients Dataset, 25 components were selected since

this is the point at which the reconstruction error falls below 0.5, indicating sufficient retention of information while achieving dimensionality reduction. For the Company Bankruptcy Dataset, 55 components were selected, as the MSE showed a smooth and linear decay, and this point provided a balance between error and reduction.

a) *Global Cancer Patients Dataset*: Figure 13 shows clustering performance before and after RP. For the cancer dataset, KMeans saw a consistent improvement in silhouette scores after RP, suggesting that RP preserved cluster structure while reducing noise. In contrast, GMM performed worse after RP, with silhouette scores dropping and turning negative in some cases. This drop may be due to the random nature of projections distorting the distributional assumptions needed by GMM. This partially supports our hypothesis: RP proved useful for enhancing KMeans clustering but not suitable for GMM, which relies on preserved Gaussian structures.

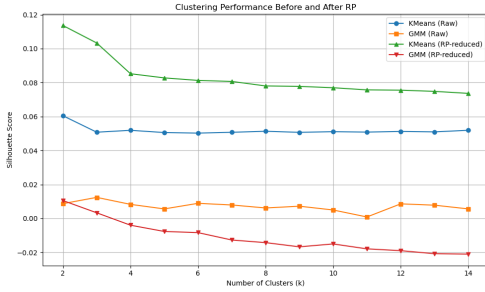


Fig. 13. Silhouette Scores: Clustering Before and After RP (Global Cancer Dataset)

b) *Company Bankruptcy Dataset*: Figure 14 shows similar trends in the bankruptcy dataset. KMeans performed better after RP for smaller values of k (2–5), indicating that RP helped emphasize separable structure early on. However, as k increased, the benefit diminished. GMM, on the other hand, initially struggled post-RP but saw slight recovery at intermediate k values before degrading again. This inconsistency reflects GMM’s sensitivity to distortions introduced by RP, especially when high-dimensional structure is compressed without preserving statistical independence. This result again partially supports our hypothesis: RP benefits KMeans by simplifying the feature space but is not robust enough to preserve the probabilistic structure needed for GMM clustering.

The combination of DR and clustering altered the geometric structure of the datasets in different ways. PCA compressed variance into fewer orthogonal directions, improving KMeans by amplifying dominant axes of separation but often degrading GMM, which relies on isotropic Gaussian assumptions. ICA reshaped the data into statistically independent components, which helped GMM most when true latent signals were present, as seen in the Cancer dataset. RP preserved pairwise distances in expectation, benefiting KMeans by preserving neighborhood relationships, but its stochastic nature distorted the density contours GMM depends on. These outcomes

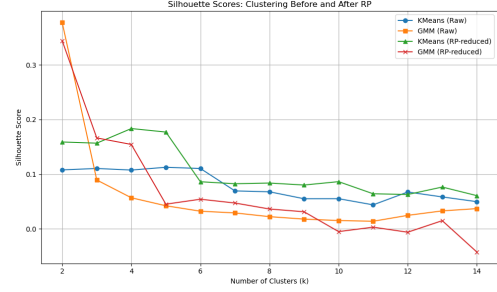


Fig. 14. Silhouette Scores: Clustering Before and After RP (Bankruptcy Dataset)

reflect how DR changes the shape, scale, and distribution of the feature space, sometimes aligning better with clustering assumptions and sometimes not.

H. Multi-Layer Perceptron Classifier on Bankruptcy Dataset

a) *Principal Component Analysis*: A Multi-Layer Perceptron (MLP) classifier was applied to the Company Bankruptcy dataset to assess the impact of dimensionality reduction via Principal Component Analysis (PCA). A fixed architecture with one hidden layer of 32 units was used for both ReLU and Tanh activations, consistent with earlier tuning results from the supervised learning experiments.

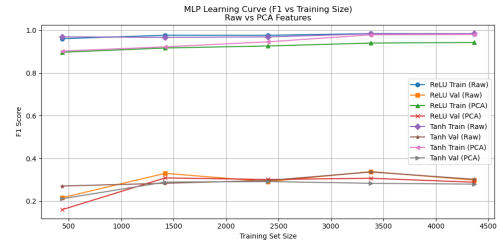


Fig. 15. MLP Learning Curve (F1 vs Training Size). Raw vs PCA Features

Figure 15 shows the F1 learning curves for both activation functions using raw and PCA-reduced inputs. For training scores, PCA significantly improved convergence for Tanh, reaching near-perfect performance with increasing training size. However, validation performance remained modest across all configurations. Specifically, Tanh with PCA achieved high training F1 but failed to generalize, indicating overfitting. ReLU showed more stable behavior, but PCA did not yield major improvements and even slightly reduced validation scores compared to raw features.

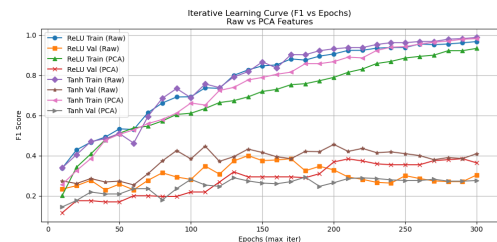


Fig. 16. Iterative Learning Curve (F1 vs Epochs). Raw vs PCA Features

Figure 16 further confirms these trends across epochs. While PCA accelerated convergence during training, particularly for Tanh, the validation curves plateaued early and failed to improve meaningfully. ReLU with PCA exhibited noisy generalization and no clear performance advantage over raw features.

Configuration	Train Time (s)	Predict Time (s)
ReLU (Raw)	3.4066	0.0005
Tanh (Raw)	4.7359	0.0019
ReLU (PCA)	2.9635	0.0004
Tanh (PCA)	4.4649	0.0018

TABLE I

MLP TRAINING AND PREDICTION TIMES: RAW VS PCA FEATURES

As shown in Table I, PCA also reduced training and prediction times slightly, further highlighting its computational benefits. The hypothesis that PCA would improve model performance and efficiency is only partially supported. PCA did lead to faster convergence and lower runtime, particularly for Tanh. However, these gains did not translate into improved generalization, and in some cases, validation performance deteriorated. This suggests that while PCA simplifies the learning problem in terms of optimization, it may also remove critical variance needed for robust prediction in imbalanced and complex datasets like this one.

b) *Multi-Layer Perceptron Classifier with ICA*: Figure 17 shows the F1 learning curves across increasing training set sizes. ReLU and Tanh models trained on ICA features begin with high F1 scores (above 90%) and show slight improvements as data size increases. Tanh with ICA achieves consistently higher validation scores than ReLU, suggesting better generalization. However, improvements over the raw feature baseline are marginal.

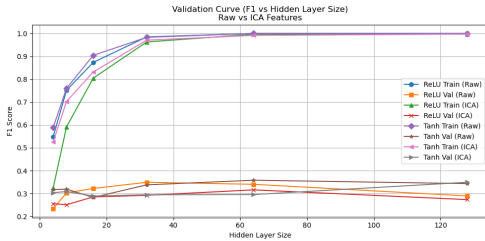


Fig. 17. MLP Learning Curve (F1 vs Training Size) Raw vs ICA Features

Figure 18 illustrates the iterative learning curve (F1 vs epochs). ICA leads to faster convergence for both activations during training, especially with Tanh. Training F1 scores reach near-perfect values ($\sim 96\text{--}98\%$) for ICA faster than with raw features. However, validation F1 performance remains limited, particularly for ReLU, and the gap between training and validation widens, indicating overfitting. This behavior suggests that ICA aids optimization but may not preserve all predictive structure in the data.

Table II summarizes model efficiency. ICA leads to modest reductions in both training and prediction times, supporting

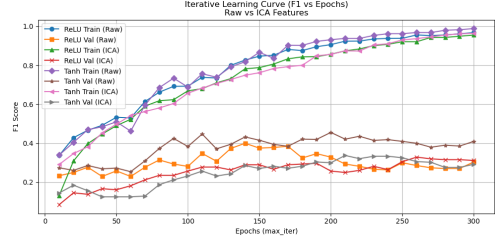


Fig. 18. Iterative Learning Curve (F1 vs Epochs) Raw vs ICA Features

the hypothesis that dimensionality reduction can improve computational performance.

TABLE II

MLP PERFORMANCE: RAW VS ICA FEATURES

Configuration	Train Time (s)	Predict Time (s)
ReLU (Raw)	3.41	0.0005
Tanh (Raw)	4.74	0.0019
ReLU (ICA)	2.60	0.0004
Tanh (ICA)	4.32	0.0021

ICA supported our hypothesis in terms of improving training speed and convergence. However, the effect on generalization performance was mixed, Tanh benefited more than ReLU, but validation scores still showed signs of overfitting. Thus, the hypothesis was only *partially supported*.

c) *Multi-Layer Perceptron Classifier with Random Projection*: Figure 19 shows the F1 learning curve against training set size. Both ReLU and Tanh activations reach high training F1 scores above 0.98 when using RP features. Validation F1 scores improve markedly with RP, especially for ReLU, which stabilizes above 0.7 across training sizes, substantially higher than its raw feature counterpart. This suggests RP enhances generalization, likely due to noise reduction and distance-preserving projections.

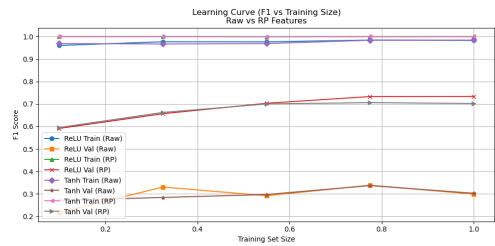


Fig. 19. MLP Learning Curve (F1 vs Training Size) Raw vs RP Features

Figure 20 illustrates iterative learning behavior over 300 epochs. Training F1 scores quickly converge near 1.0 for all models. However, validation performance diverges. ReLU with RP maintains consistently high F1 scores throughout training, outperforming all other configurations. Tanh with RP also improves over its raw counterpart but shows a slight decline in validation scores at later epochs, hinting at mild overfitting.

Table III summarizes training and prediction time. RP substantially reduces training time, by over 75% for ReLU and

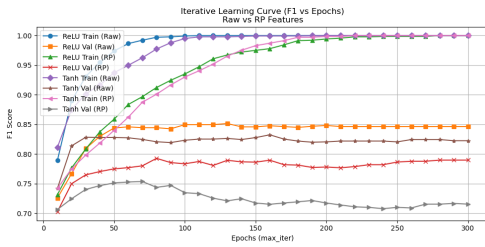


Fig. 20. Iterative Learning Curve (F1 vs Epochs) Raw vs RP Features

77% for Tanh, demonstrating a key computational advantage of dimensionality reduction without sacrificing performance.

TABLE III
MLP RUNTIME PERFORMANCE: RAW VS RP FEATURES

Model	Train Time (s)	Predict Time (s)
ReLU (Raw)	3.4066	0.0005
Tanh (Raw)	4.7359	0.0019
ReLU (RP)	0.7267	0.0004
Tanh (RP)	1.0778	0.0010

Dimensionality reduction influenced the neural network's training dynamics in several ways. PCA and ICA accelerated convergence by simplifying the input space, but often at the cost of generalization, especially with Tanh, which showed sharper overfitting and earlier plateauing. This suggests that while DR lowers the input dimensionality, it may also reduce the model's effective capacity to capture subtle decision boundaries, especially in imbalanced datasets like Bankruptcy. RP, in contrast, preserved distance relationships more robustly and yielded better generalization with ReLU, as evidenced by smoother validation curves and smaller train-val gaps. These patterns show that the interplay between DR geometry and model architecture (activation function, capacity) significantly shapes learning outcomes.

d) *Extra Credit*: Isomap was also applied to project data into a lower-dimensional manifold. The results were mixed. While training time was significantly reduced, validation F1-scores dropped relative to PCA and RP. ReLU with Isomap achieved a validation F1 of 0.65, which is promising, but still below RP. Tanh underperformed on validation despite good recall, highlighting potential instability in the optimization landscape when using non-linear transformations on high-dimensional imbalanced data.

IV. CONCLUSION

In this project, we evaluated the impact of unsupervised learning techniques, clustering (K-Means, EM) and dimensionality reduction (PCA, ICA, RP, Isomap), on two real-world datasets: Global Cancer Staging and Company Bankruptcy. We also assessed how these techniques influence downstream supervised learning performance using a Multi-Layer Perceptron (MLP) classifier.

Our hypothesis had three components: that dimensionality reduction would improve clustering quality, reduce neural

TABLE IV
NEURAL NETWORK PERFORMANCE ACROSS DIMENSIONALITY
REDUCTION METHODS (BANKRUPTCY DATASET)

Model	Train Time (s)	Predict Time (s)	F1-Score	Recall
ReLU (Raw)	3.41	0.0005	0.9574	0.0728
Tanh (Raw)	4.74	0.0019	0.9630	0.3510
ReLU (PCA)	2.96	0.0004	0.9584	0.2727
Tanh (PCA)	4.46	0.0018	0.9541	0.2955
ReLU (ICA)	2.60	0.0004	0.9584	0.2727
Tanh (ICA)	4.32	0.0021	0.9541	0.2955
ReLU (RP)	0.73	0.0004	0.7796	0.8212
Tanh (RP)	1.08	0.0010	0.7150	0.7119
ReLU (Isomap)	0.64	0.0003	0.6532	0.6358
Tanh (Isomap)	0.96	0.0009	0.6029	0.6358

network training time, and preserve or improve classification performance.

Clustering performance improved in many cases post-reduction, particularly for KMeans following PCA or RP. ICA showed strong gains in select configurations, especially where non-Gaussian latent structure was present. GMM generally underperformed after DR, supporting the idea that the success of clustering post-reduction depends on alignment between DR geometry and clustering assumptions. **Thus, the first part of the hypothesis was partially supported.**

Training efficiency consistently improved across all DR techniques, particularly with RP, which yielded substantial speedups (up to 75%) without sacrificing performance. **This strongly supports the second part of the hypothesis.**

Classification performance was mixed: PCA and ICA often led to faster convergence but also overfitting or reduced generalization, especially with Tanh. Only RP yielded stable improvements in validation F1 scores. **Therefore, the third part of the hypothesis was partially supported, with RP being the most effective.**

Overall, dimensionality reduction proved useful, but its impact varied by method, dataset, and task. The results reaffirm the importance of aligning DR methods with downstream algorithmic assumptions, especially in high-dimensional and imbalanced settings.

REFERENCES

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